网络安全实验

防火墙

防火墙实验

01

访问控制部分

- 设置iptables规则，实现指定功能
- 设置iptables规则，实现特定远端客户SSH连接

02

NAT部分

- 设置iptables规则，实现内网服务的发布

01

访问控制部分

- 设置iptables规则，实现指定功能
- 设置iptables规则，实现特定远端客户SSH连接

iptables

防火墙

Kali

其他主机

Ubuntu

软件

iptables、wireshark

查看主机信息

使用命令ifconfig查看主机地址信息。两虚拟机之间应能互相ping通。

```
┌──(root㉿kali)-[~]
└─# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.133.136  netmask 255.255.255.0  broadcast 192.168.133.255
        inet6 fe80::94e4:385a:c603:9d16  prefixlen 64  scopeid 0×20<link>
        ether 00:0c:29:96:4a:12  txqueuelen 1000  (Ethernet)
        RX packets 322  bytes 50277 (49.0 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 301  bytes 46240 (45.1 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0×10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 4  bytes 240 (240.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 4  bytes 240 (240.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0


┌──(root㉿kali)-[~]
└─# ping 192.168.133.134
PING 192.168.133.134 (192.168.133.134) 56(84) bytes of data.
64 bytes from 192.168.133.134: icmp_seq=1 ttl=64 time=1.46 ms
64 bytes from 192.168.133.134: icmp_seq=2 ttl=64 time=0.521 ms
64 bytes from 192.168.133.134: icmp_seq=3 ttl=64 time=0.500 ms
^C
── 192.168.133.134 ping statistics ──
3 packets transmitted, 3 received, 0% packet loss, time 2018ms
rtt min/avg/max/mdev = 0.500/0.826/1.458/0.446 ms
```

```
ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.133.134  netmask 255.255.255.0  broadcast 192.168.133.255
        inet6 fe80::faed:fcf8:4ed:4207  prefixlen 64  scopeid 0x20<link>
        ether 00:0c:29:f0:06:e4  txqueuelen 1000  (Ethernet)
        RX packets 286530  bytes 375603684 (375.6 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 66044  bytes 5747615 (5.7 MB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 23847  bytes 2075951 (2.0 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 23847  bytes 2075951 (2.0 MB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

ubuntu@ubuntu-virtual-machine:~$ ping 192.168.133.136
PING 192.168.133.136 (192.168.133.136) 56(84) bytes of data.
64 bytes from 192.168.133.136: icmp_seq=1 ttl=64 time=3.53 ms
64 bytes from 192.168.133.136: icmp_seq=2 ttl=64 time=0.470 ms
64 bytes from 192.168.133.136: icmp_seq=3 ttl=64 time=0.435 ms
^C
--- 192.168.133.136 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2008ms
rtt min/avg/max/mdev = 0.435/1.478/3.529/1.450 ms
```

iptables

禁止所有主机ping本地主机

命令示例：iptables -A INPUT -p icmp -j DROP

```
┌─(root㉿kali)-[~]
└─# iptables -A INPUT -p icmp -j DROP
```

```
ubuntu@ubuntu-virtual-machine:~$ ping 192.168.133.136
PING 192.168.133.136 (192.168.133.136) 56(84) bytes of data.
64 bytes from 192.168.133.136: icmp_seq=1 ttl=64 time=3.53 ms
64 bytes from 192.168.133.136: icmp_seq=2 ttl=64 time=0.470 ms
64 bytes from 192.168.133.136: icmp_seq=3 ttl=64 time=0.435 ms
^C
--- 192.168.133.136 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2008ms
rtt min/avg/max/mdev = 0.435/1.478/3.529/1.450 ms
ubuntu@ubuntu-virtual-machine:~$ ping 192.168.133.136
PING 192.168.133.136 (192.168.133.136) 56(84) bytes of data.
^C
--- 192.168.133.136 ping statistics ---
8 packets transmitted, 0 received, 100% packet loss, time 7174ms
```

可以ping通

不能ping通

仅允许某特定IP主机ping本地主机

命令示例：iptables -I INPUT -p icmp -s 192.168.133.134 -j ACCEPT

```
┌─(root㉿kali)-[~]
└─# iptables -L
Chain INPUT (policy ACCEPT)
target     prot opt source               destination
ACCEPT     icmp --  192.168.133.134      anywhere
DROP       icmp --  anywhere             anywhere

Chain FORWARD (policy ACCEPT)
target     prot opt source               destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source               destination
```

```
ubuntu@ubuntu-virtual-machine:~$ ping 192.168.133.136
PING 192.168.133.136 (192.168.133.136) 56(84) bytes of data.
^C
--- 192.168.133.136 ping statistics ---
8 packets transmitted, 0 received, 100% packet loss, time 7174ms

ubuntu@ubuntu-virtual-machine:~$ ping 192.168.133.136
PING 192.168.133.136 (192.168.133.136) 56(84) bytes of data.
64 bytes from 192.168.133.136: icmp_seq=1 ttl=64 time=0.447 ms
64 bytes from 192.168.133.136: icmp_seq=2 ttl=64 time=0.451 ms
64 bytes from 192.168.133.136: icmp_seq=3 ttl=64 time=0.421 ms
^C
--- 192.168.133.136 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2044ms
rtt min/avg/max/mdev = 0.421/0.439/0.451/0.013 ms
```

不能ping通

再次ping通

iptables

允许每10s通过一个ping包

命令示例：iptables -I INPUT -p icmp -m limit --limit 6/min --limit-burst 1 -j ACCEPT

iptables -A INPUT -p icmp -j DROP

```
┌──(root㉿kali)-[~]
└─# iptables -L
Chain INPUT (policy ACCEPT)
target     prot opt source               destination
ACCEPT     icmp --  anywhere             anywhere             limit: avg 6/min burst 1
DROP       icmp --  anywhere             anywhere

Chain FORWARD (policy ACCEPT)
target     prot opt source               destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source               destination
```

```
ubuntu@ubuntu-virtual-machine:~$ ping 192.168.133.136
PING 192.168.133.136 (192.168.133.136) 56(84) bytes of data.
64 bytes from 192.168.133.136: icmp_seq=1 ttl=64 time=0.704 ms
64 bytes from 192.168.133.136: icmp_seq=11 ttl=64 time=0.603 ms
64 bytes from 192.168.133.136: icmp_seq=21 ttl=64 time=0.555 ms
^C
--- 192.168.133.136 ping statistics ---
21 packets transmitted, 3 received, 85.7143% packet loss, time 20480ms
rtt min/avg/max/mdev = 0.555/0.620/0.704/0.062 ms
```

阻断来自某MAC地址的数据包

命令示例：iptables -A INPUT -m mac --mac-source <your_MAC> -j DROP

```
┌──(root㉿kali)-[~]
└─# iptables -A INPUT -m mac --mac-source 00:0c:29:f0:06:e4 -j DROP

┌──(root㉿kali)-[~]
└─# iptables -L
Chain INPUT (policy ACCEPT)
target     prot opt source               destination
DROP       all  --  anywhere             anywhere             MAC 00:0c:29:f0:06:e4

Chain FORWARD (policy ACCEPT)
target     prot opt source               destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source               destination
```

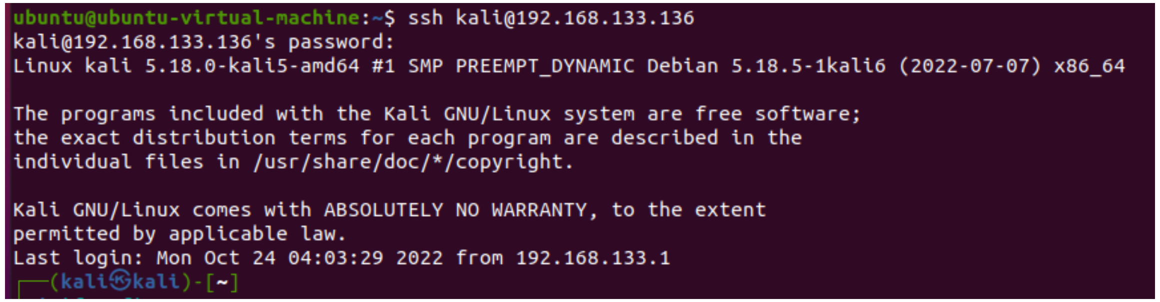
```
ubuntu@ubuntu-virtual-machine:~$ ping 192.168.133.136
PING 192.168.133.136 (192.168.133.136) 56(84) bytes of data.
^C
--- 192.168.133.136 ping statistics ---
6 packets transmitted, 0 received, 100% packet loss, time 5106ms
```

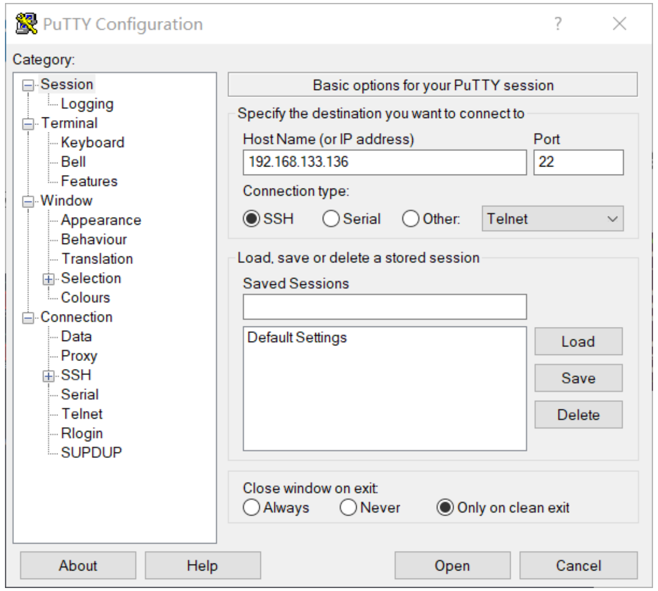
允许特定主机SSH连接

启动服务器的SSH服务

启动服务，使得客户端可以通过22端口远程连接

客户端1、2分别SSH连接服务器





允许特定主机SSH连接

设置服务器的iptables规则

设置iptables规则，只允许特定远端主机SSH连接。

命令示例：iptables -I INPUT -p tcp --dport 22 -s 192.168.133.134 -j ACCEPT

iptables -A INPUT -p tcp --dport 22 -j DROP

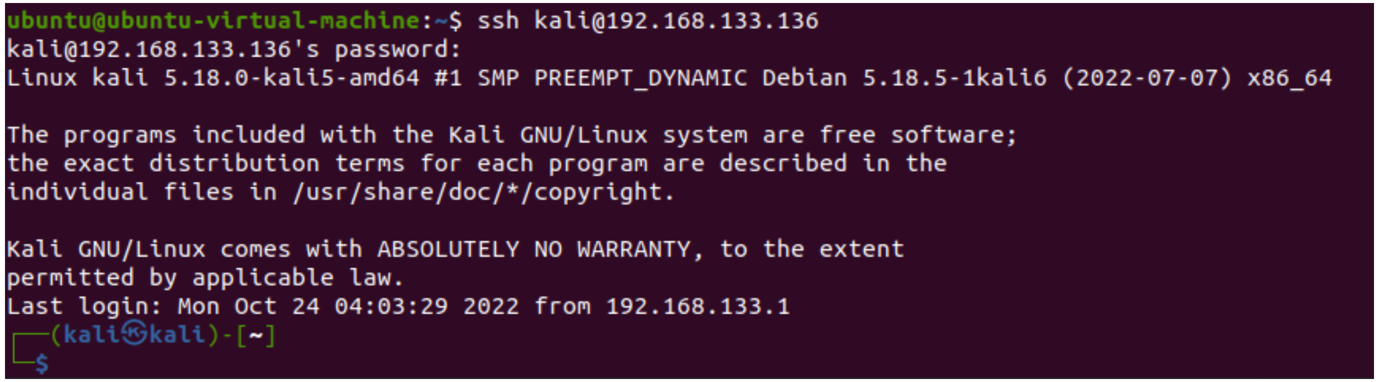
```
┌──(root㉿kali)-[~]
└─# iptables -L
Chain INPUT (policy ACCEPT)
target     prot opt source               destination
ACCEPT     tcp  --  192.168.133.134      anywhere             tcp dpt:ssh
DROP       tcp  --  anywhere             anywhere             tcp dpt:ssh

Chain FORWARD (policy ACCEPT)
target     prot opt source               destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source               destination
```

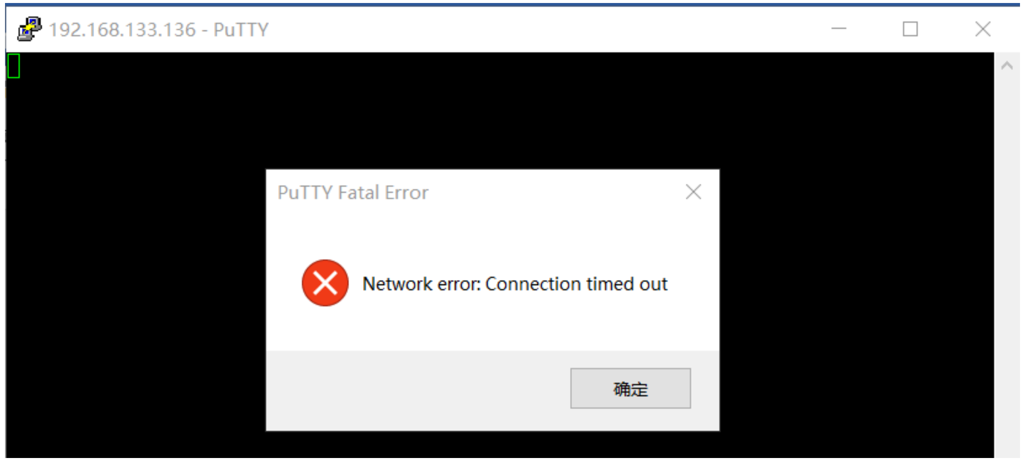
允许特定主机SSH连接

验证防火墙效果

在iptables中被允许连接的客户端1仍可远程SSH连接

客户端2已无法连接到服务器





02

NAT部分

- 设置iptables规则，实现内网http服务的发布

发布内网服务

防火墙

Ubuntu

内网主机PC1

Windows XP

公网主机PC2

Windows 10

软件

iptables、nginx

了解实验场景

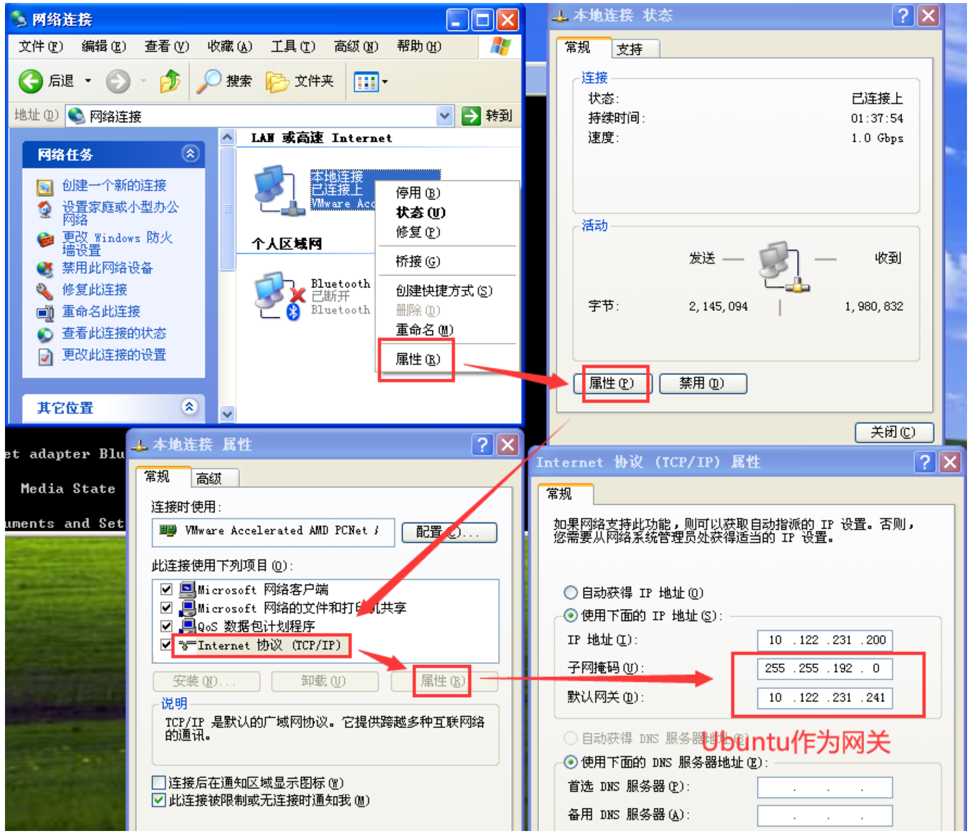
PC2不能直接访问局域网内的电脑PC1上的服务，但PC2可以与firewall上的公网IP相互通讯，由此可以进行NAT的设置，让PC2访问firewall的公网地址上的服务，再由firewall对数据进行处理，将访问请求发送到内网PC1上。

发布内网服务

配置网络环境

给作为防火墙的主机增设网卡，两张网卡使用不同的网络连接方式。

PC1和PC2分别与防火墙的两张网卡的连接方式相对应，同时将PC1的网关设置为部署了防火墙的主机。



发布内网服务

测试网络连通性

完成网络环境设置后，应实现：

- PC1可以和防火墙互相ping通；
- PC2可以和防火墙互相ping通；
- PC1和PC2互ping不通

开启防火墙的路由转发，设置NAT规则

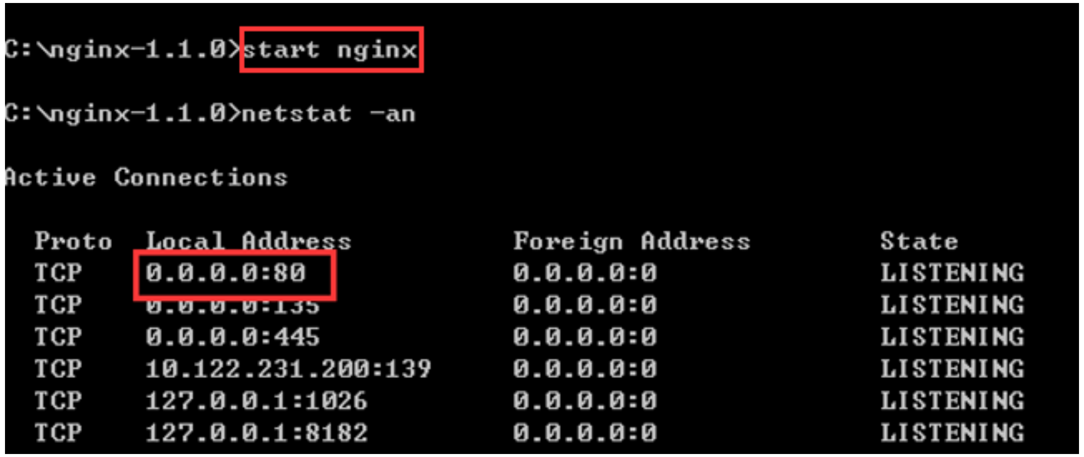
命令示例：

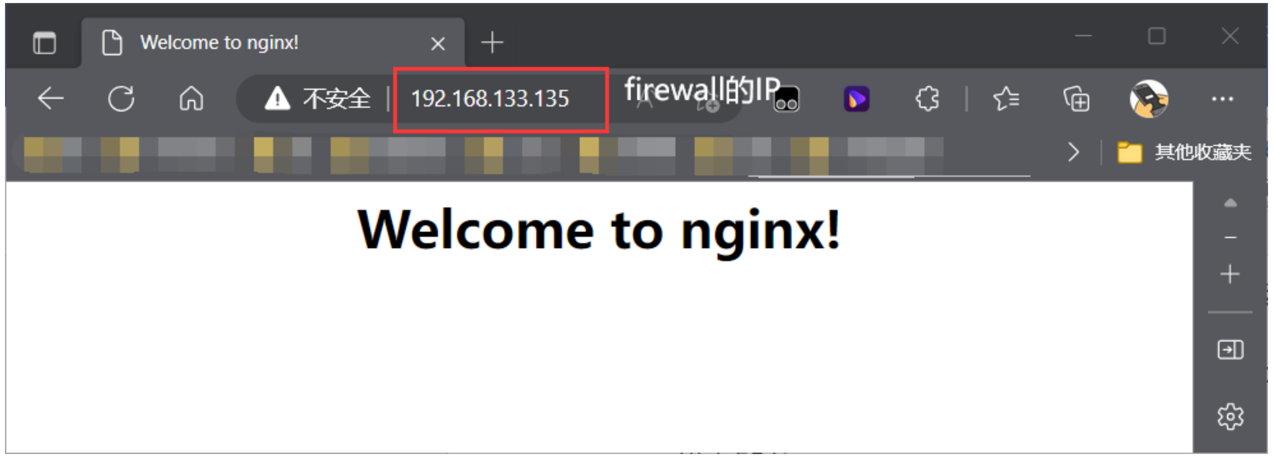
```
echo 1 > /proc/sys/net/ipv4/ip_forward
iptables -t nat -A PREROUTING -i ens38 -p tcp -d 192.168.133.135 --dport 80 -j DNAT --to-destination 10.122.231.200
iptables -t nat -A POSTROUTING -o ens38 -p tcp -j SNAT --to-source 192.168.133.135
iptables -A FORWARD -i ens38 -p tcp -d 10.122.231.200 --dport 80 -j ACCEPT
iptables -A FORWARD -p tcp -m state --state established,related -j ACCEPT
```

发布内网服务

公网主机PC2使用防火墙IP访问内网服务

启动PC1的http服务后，PC2可以通过使用防火墙公网IP访问内网服务





谢谢！